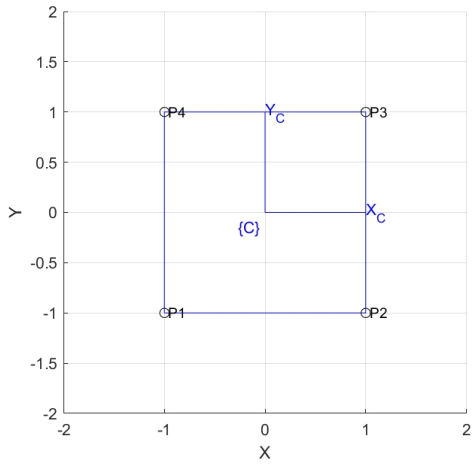
Lab 1 – Coordinate Transforms (2D)

1 Introduction

In this lab the mathematics behind 2D coordinate systems and transformation between coordinate frames will be exercised. MATLAB will be used initially to become familiar with the operations and setup of coordinate transforms.

2 2D Translation and Rotation

The world coordinate system is defined as {W}. A 2x2 square object is described by four points. The origin of coordinate system {C} is located at the center of the square as shown.



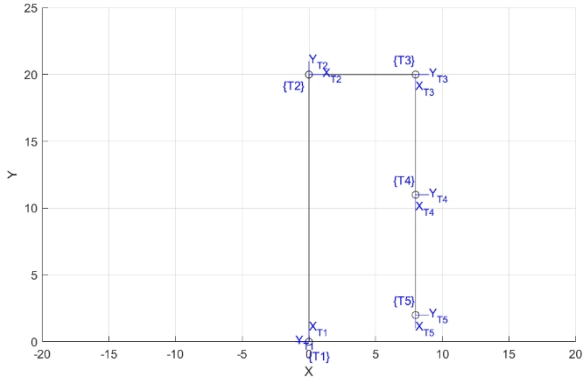
1) Create a homogeneous transformation to place {C} at location (4,3) with a rotation of +45 degrees. Plot {W}, {C} and the four corners of the square.

2) Using a FOR loop, rotate the square 360 degrees about the origin in {C}. Show resulting animation using the `drawnow` command inside the loop (use a step size so it takes approximately 4 seconds).

3) Using a FOR loop, rotate the square 360 degrees about the origin in {W}. Show resulting animation (use a step size so it takes approximately 4 seconds).

3 Planar Robot

1) Measure 5 translations on yourself with respect to the world coordinate system {W} at the center of waist, neck {B}, shoulder {C}, elbow {D} and wrist {E}. Create a set of 5 points at the origin of each coordinate system i.e. ${}^{E}P=(0,0)$, ${}^{D}P=(0,0)$, etc. Transform each of these points to the world coordinate system using cascading transformations and plot on the display. For example the point ${}^{C}P$ is transformed to the world coordinate system as follows: ${}^{W}P = {}^{W}T_{B} * {}^{B}T_{C} * {}^{C}P$.



2) Apply a rotation from +0 to +45 degrees to the transformation at frame {C} the shoulder in a FOR loop. Plot the robot body while slowly incrementing the angle. Use the HOLD ON command in MATLAB. Verify the elbow and wrist move appropriately.

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Example code to plot a square:
P1 = [-1, -1]';
P2 = [1, -1]';
P3 = [1, 1]';
P4 = [-1, 1]';
pts=[(P1)';(P2)';(P3)';(P4)';(P1)'];
plotvol([-5 5 -5 5])
plot (pts(:,1),pts(:,2),'b-');
plot_point(P1(1:2), 'label', 'P1', 'ko')
plot_point(P2(1:2), 'label', 'P2', 'ko')
plot_point(P3(1:2), 'label', 'P3', 'ko')
plot_point(P4(1:2), 'label', 'P4', 'ko')
```